

Question # 1:

[8 x 2 = 16 Marks]

a) Solution:

"The device overheats unless the fan is running" $\neg F \rightarrow D$

"The fan will run only if the voltage is stable" $F \rightarrow V$

"The device will not overheat if the voltage is stable" $V \rightarrow \neg D$

Full statement: $[(\neg F \rightarrow D) \wedge (F \rightarrow V)] \rightarrow (V \rightarrow \neg D)$

b) Solution:

Symbolically: Original statement: $(D \wedge F) \rightarrow V$; Contrapositive: $\neg V \rightarrow \neg (D \wedge F)$ OR $\neg V \rightarrow (\neg D \vee \neg F)$

Contrapositive: "If the voltage is stable, then it is not the case that the device overheats and the fan is not running." OR "If the voltage is stable then either the device doesn't overheat or the fan is running."

Apply De Morgan: $\neg (V \vee (\neg D \vee \neg F)) \equiv \neg V \wedge (D \wedge F)$

Statement: "The voltage is stable and the device overheats and the fan is not running."

c) Solution:

The alarm will sound(A) if the front door is unlocked(U). $U \rightarrow A$

The alarm sounds(A) only if the security company is notified(N). $A \rightarrow N$

The police are called(P) when the security company is notified(N). $N \rightarrow P$

The front door is unlocked(U).

Apply Modus Ponens repeatedly

1. From $U \rightarrow A$ and U, we get A.

2. From $A \rightarrow N$ and A, we get N.

3. From $N \rightarrow P$ and N, we get P.

d) i) Solution: Not everyone in your class has an Internet connection.

ii) Solution: Someone in your class has an Internet connection but has not chatted with anyone else in your class. $\exists$

e) i) Solution: $\forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (x + y < 0))$ ii) Solution: $\neg \forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x - y > 0))$

f) i) Solution: $\{1, 1, 1, 1, 0, 0, 0\}$

ii) Solution: $A - B = \{(CS, 2), (AI, 1), (CY, 1)\}$

g) Solution:

$A \cap B = \{x \mid x \text{ is a multiple of 6}\}$

$A \cup B = \{x \mid x \text{ is a multiple of 2 or 3}\}$

$A - B = \{x \mid x \text{ is a multiple of 2 but not a multiple of 3}\}$

h) i) Solution: Yes. f can be onto because every course in C can have at least one student assigned. For example: $f(s_1) = c_1, f(s_2) = c_1, f(s_3) = c_2, f(s_4) = c_3$. All courses c_1, c_2, c_3 have at least one student $\rightarrow f$ is onto.

ii) Solution: No. f cannot be one-to-one because $|S| = 4$ and $|C| = 3$. A one-to-one function requires each element of S to map to a distinct element of C . Since there are more students than courses, at least two students must share a course. Therefore, f cannot be one-to-one.

Question # 2:

[8 x 2 = 16 Marks]

a) Solution:

Solution

For surjectivity, notice that the range, and co-domain is, $\mathbb{Z}$, the set of all real numbers. Also for any integer, m , $f(m) = \lfloor m \rfloor = m$, meaning that every integer, m , is reached, or obtained by the floor function, $f(x) = \lfloor x \rfloor$.

For injectivity, notice that multiple numbers are rounded down to the same integer. For example, $f(3.4) = \lfloor 3.4 \rfloor = 3$, and $f(3.7) = \lfloor 3.7 \rfloor = 3$, so that $f(3.4) = f(3.7)$, but $3.4 \neq 3.7$

b) Solution:

The "beats" relation is not transitive because there could exist a cycle of length 3 where x beats y, y beats z and z beats x. By the definition of a tournament, x cannot then beat y in such a situation. The "beats" relation is antisymmetric since for any distinct players x and y, if x beats y then y does not beat x. The "beats" relation is not reflexive since a tournament graph has no self-loops.

c) Solution:

i) This is not a tree since it contains a cycle. ii) This is not a tree since it is not connected.

d) Solution:

of edges: $e = 30$; # of regions: $r = 20$; # of vertices: $v = ?$
Euler formula: $v = e - r + 2 = 30 - 20 + 2 = 12$.

e) Solution:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

f) Solution: Table # 01: Dijkstra's shortest-path algorithm computation

N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)
a	10,a	7,a	-	-	6,a	-
af	9,f	7,a	-	8,f		9,f
afc	9,f		14,c	8,f		9,f
afce	9,f		14,c			9,f
afceb			12,b			9,f
afcebg			10,g			
afcebgd						

g) Solution:

Order of edges: $(g, d) = 1$, $(g, f) = 3$, $(f, e) = 2$, $(d, b) = 3$, $(b, c) = 2$, $(a, f) = 6$. MST Cost = 17

h) Solution:

Pre order: STOP, LET, ANT, NONE, NINE, OAT, THERE, TAPE, YOU, ZOO.
In order: ANT, LET, NINE, NONE, OAT, STOP, TAPE, THERE, YOU, ZOO.

Question # 3:

[8 x 2 = 16 Marks]

a) Solution:

First, we find the A.P with the common difference $(d) = 7$

259, 266, 273, 280,, 784

Since $T_n = a + (n-1)d$; $784 = 259 + (n-1)(7)$; $n = 76$.

Now for Sum; $S_n = n/2 [2a + (n-1)d]$; $S_{76} = 76/2 [2(259) + (76-1)(7)] = 39,634$

b) Solution:

Here $a = 1250$, $T_n = 10000$, $n = 4$, $r = ?$

$T_n = ar^{n-1}$; $10000 = 1250 r^{4-1}$;

$\frac{10000}{1250} = r^3$;

$r = 2$. Hence the rate of increase in bacteria is 200%.

c) Solution:

Proof:

1. Suppose a is any odd integer and b is any even integer. [We must show that $a - b$ is odd.]
2. By definition of odd, $a = 2r + 1$ for some integer r , and $b = 2s$ for some integer s .
3. Then $a - b = (2r + 1) - 2s$ by substitution
4. $= 2r - 2s + 1$ by combining like terms
5. $= 2(r - s) + 1$ by factoring out 2.
6. Let $t = r - s$.
7. Then t is an integer because it is a difference of integers.
8. So, by substitution, $a - b = 2t + 1$, where t is an integer.
9. Hence $a - b$ is odd [as was to be shown].

d) Solution:

Contrapositive: If $a \geq 2$ then $a^2 - 4b^2 \geq 0$.

Proof: If $a \geq 2b$, square both sides (both nonnegative):

$$a^2 \geq 4b^2 \Rightarrow a^2 - 4b^2 \geq 0.$$

Thus, the contrapositive is true, so the original statement holds.

e) Solution:

Assume n is divisible by 2 but $n+1$ is not odd. Show this assumption leads to a contradiction.

Since n is divisible by 2, hence $n = 2k$

Now, $n + 1 = 2k + 1$ which is odd, hence our supposition is wrong.

f) Solution: For $p=2$, $p^2 - 1 = (2)^2 - 1 = 3$.

g) Solution:

SOLUTION:

Let $P(n)$ be the given equation.

1. Basis Step:

$P(1)$ is true

For $n = 1$

$$\text{L.H.S of } P(1) = \frac{1}{1 \cdot 2} = \frac{1}{1 \times 2} = \frac{1}{2}$$

$$\text{R.H.S of } P(1) = \frac{1}{1+1} = \frac{1}{2}$$

Hence $P(1)$ is true

2. Inductive Step:

Suppose $P(k)$ is true, for some integer $k \geq 1$. That is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k(1)}{k+1}$$

To prove $P(k+1)$ is true. That is

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+1+1)} = \frac{k+1}{(k+1)+1} \dots \dots (2)$$

Now we will consider the L.H.S of the equation (2) and will try to get the R.H.S by using equation (1) and some simple computation.

Consider LHS of (2)

$$\begin{aligned} & \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+2)} \\ &= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2)+1}{(k+1)(k+2)} \\ &= \frac{k^2+2k+1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{k+1}{k+2} \\ &= \text{RHS of (2)} \end{aligned}$$

Hence $P(k+1)$ is also true and so by Mathematical induction the given equation is true for all integers $n \geq 1$.

h) Solution:

$$\begin{aligned} (3x - y^2)^4 &= \sum_{k=0}^4 \binom{4}{k} (3x)^{4-k} (-y^2)^k \\ (3x - y^2)^4 &= \binom{4}{0} (3x)^4 (-y^2)^0 + \binom{4}{1} (3x)^3 (-y^2)^1 + \binom{4}{2} (3x)^2 (-y^2)^2 + \binom{4}{3} (3x)^1 (-y^2)^3 + \binom{4}{4} (3x)^0 (-y^2)^4 \\ &= 81x^4 + 108x^3(-y^2) + 54x^2y^4 + 12x(-y^6) + y^8 \\ &= 81x^4 - 108x^3y^2 + 54x^2y^4 - 12xy^6 + y^8 \end{aligned}$$

Question # 4:

[6 x 2 + 4 = 16 Marks]

a) Solution: Given $a = 4146$ and $b = 1989$

$$4146 = 2 \cdot 1989 + 168;$$

$$1989 = 11 \cdot 168 + 141;$$

$$168 = 1 \cdot 141 + 27;$$

$$141 = 5 \cdot 27 + 6;$$

$$27 = 4 \cdot 6 + 3;$$

$$6 = 2 \cdot 3 + 0$$

By backward substitution,

$$3 = 4146 \cdot 296 + 1989 \cdot (-617).$$

Hence, $x = 296$, and $y = -617$.

b) Solution:

$$C(S) = 18^{13} \bmod 2537$$

$$C(O) = 14^{13} \bmod 2537$$

$$C(L) = 11^{13} \bmod 2537$$

$$C(I) = 08^{13} \bmod 2537$$

$$C(D) = 03^{13} \bmod 2537$$

c) Solution:

Digits $d_1 \dots d_9 = 0, 3, 0, 6, 4, 0, 6, 1, 5$. Need $\sum_{i=1}^{10} id_i \equiv 0 \pmod{11}$.

Compute $\sum_{i=1}^9 id_i = 145$. Solve $145 + 10d_{10} \equiv 0 \pmod{11}$. This gives $10d_{10} \equiv 9 \pmod{11} \Rightarrow d_{10} \equiv 2 \pmod{11}$.

d) Solution:

1. Reduce the exponent modulo 18:

$$1234 \div 18 = 68 \text{ remainder } 10, \text{ so } 1234 \equiv 10 \pmod{18}$$

Thus $17^{1234} \equiv 17^{10} \pmod{19}$.

2. Simplify the base: $17 \equiv -2 \pmod{19}$. So

$$17^{10} \equiv (-2)^{10} = 2^{10} = 1024 \pmod{19}.$$

3. Reduce 1024 modulo 19:

$$19 \cdot 53 = 1007, \quad 1024 - 1007 = 17.$$

Therefore

$$\boxed{17^{1234} \equiv 17 \pmod{19}.$$

- e) i) Solution: Use $aa^{-1} = 1 \pmod{m}$; hence, $34 \times 55 = 1 \pmod{89}$;
 ii) Solution: Use $aa^{-1} = 1 \pmod{m}$; hence, $73 \times 89 = 1 \pmod{232}$.

f) Solution: Encrypted Message: "STOP TALKING".

g) Solution: $x \equiv 7 \pmod{11}$, $x \equiv 4 \pmod{15}$, $x \equiv 2 \pmod{7}$,

1. Compute the product of all moduli:

$$M = 11 \cdot 15 \cdot 7 = 1155.$$

2. For each congruence $x \equiv a_i \pmod{m_i}$ form $M_i = M/m_i$:

$$M_1 = \frac{1155}{11} = 105, \quad M_2 = \frac{1155}{15} = 77, \quad M_3 = \frac{1155}{7} = 165$$

3. Find y_i such that $M_i y_i \equiv 1 \pmod{m_i}$ (modular inverse of M_i modulo m_i).

- For $m_1 = 11$: $105 \pmod{11} = 6$. Solve $6y_1 \equiv 1 \pmod{11}$.
 $y_1 = 2$ because $6 \cdot 2 = 12 \equiv 1 \pmod{11}$.
- For $m_2 = 15$: $77 \pmod{15} = 2$. Solve $2y_2 \equiv 1 \pmod{15}$.
 $y_2 = 8$ because $2 \cdot 8 = 16 \equiv 1 \pmod{15}$.
- For $m_3 = 7$: $165 \pmod{7} = 4$. Solve $4y_3 \equiv 1 \pmod{7}$.
 $y_3 = 2$ because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$.

4. Form the CRT sum

$$x \equiv \sum_{i=1}^3 a_i M_i y_i \pmod{M}.$$

Plug in $a_1 = 7$, $a_2 = 4$, $a_3 = 2$:

$$x \equiv 7 \cdot 105 \cdot 2 + 4 \cdot 77 \cdot 8 + 2 \cdot 165 \cdot 2 \pmod{1155}.$$

5. Reduce modulo 1155:

$$4594 \pmod{1155} = 4594 - 3 \cdot 1155 = 4594 - 3465 = 1129.$$

So one solution is $x = 1129$. Since CRT solutions are unique modulo M ,

$$x \equiv 1129 \pmod{1155}$$

and the smallest positive solution is **1129**.

Question # 5:

[6 x 2 + 4 = 16 Marks]

a) Solution:

There are integers from -10 to 40 inclusive. That gives $40 - (-10) + 1 = 51$ possible integer temperatures (these are the pigeonholes). The 365 daily readings are the pigeons. Because, $365 > 51$, you cannot place each day into a distinct temperature slot — by the pigeonhole principle at least one temperature slot must contain at least two days. Therefore two (or more) days have exactly the same recorded temperature. As one temperature must appear at least $\lceil 365/51 \rceil$ times. Since $51 \cdot 7 = 357$ and $365 - 357 = 8$, so some temperature occurs on at least 8 days.

b) i) Solution: Each question has three possible responses: True, False, and Blank. So, each question = 3 choices. There are 100 questions, each answered independently. Thus, 3^{100} .

ii) Solution: A student may also leave a question blank. So, for each question, the student has: 4 choices + 1 blank option = 5 possible responses. Each of the 20 questions can be answered in 5 ways, independently. Thus, 5^{20} .

c) Solution:

Solution. There are $\binom{8}{5} = 56$ possible different plans (the pigeonholes). If there are n students, then from the pigeon-holes principle there will be at least $\lceil n/56 \rceil$ students with the same plan, and the least n such that $\lceil n/56 \rceil \geq 10$ is $n = 9 \cdot 56 + 1 = 505$.

d) Solution:

We choose 2 shirts out of 5 for both Amy and Bunny to wear, so $\binom{5}{2} 2! = 20$.

We choose 2 pairs of pants out of 4 for them to wear, so $\binom{4}{2} 2! = 12$.

We choose 2 pairs of shoes out of 2 for them to wear, so $\binom{2}{2} 2! = 2$.

Therefore, by the rule of product, the answer is $20 \times 12 \times 2 = 480$ ways. $\square$

e) Solution: Coefficient = $(^{17}C_{10}) (3)^7 (2)^{10} = 4.355 \times 10^{10}$.

f) Solution:

Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8. By the sum rule $P = P_6 + P_7 + P_8$. Finding P_6 directly is difficult. To find P_6 it is easier to find the number of strings of uppercase letters and digits that are six characters long, including those with no digits, and subtract from this the number of strings with no digits. By the product rule, the number of strings of six characters is 36^6 , and the number of strings with no digits is 26^6 .

To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

- $P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560$.
- $P_7 = 36^7 - 26^7 = 78,364,164,096 - 8,031,810,176 = 70,332,353,920$.
- $P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880$.

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

g) Solution:

For a three-letter block $L_1 L_2 L_3$ its numeric value is

$$V(L_1 L_2 L_3) = 26^2 \cdot L_1 + 26 \cdot L_2 + L_3.$$

Compute values:

- For BHY:

$$B = 1, H = 7, Y = 24 \text{ so}$$

$$V(BHY) = 676 \cdot 1 + 26 \cdot 7 + 24 = 676 + 182 + 24 = 882.$$

- For CDX:

$$C = 2, D = 3, X = 23 \text{ so}$$

$$V(CDX) = 676 \cdot 2 + 26 \cdot 3 + 23 = 1352 + 78 + 23 = 1453.$$

Step 2 — Serial number of a full plate

Each letter-block has 1000 numeric plates (000–999).

So the serial number for a plate ABC ddd is

$$\text{Serial} = V(ABC) \times 1000 + (\text{digits}).$$

Compute serials:

- BHY 026:

$$\text{Serial}_{\text{start}} = 882 \times 1000 + 26 = 882,026.$$

- CDX 100:

$$\text{Serial}_{\text{end}} = 1453 \times 1000 + 100 = 1,453,100.$$

Step 3 — Count inclusive

Number issued from start up to and including end:

$$\text{Count} = \text{Serial}_{\text{end}} - \text{Serial}_{\text{start}} + 1.$$

Plug in:

$$\text{Count} = 1,453,100 - 882,026 + 1 = 571,075$$

Final answer

571,075 license plates

were issued from BHY 026 up to and including CDX 100.